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Collected Mathematical Papers

Volume IX, Pages 529–540 (Papers 617–619)

Transcribed from the Original Publications

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617. On the Scalene Transformation of a Plane Curve (continued)

[p. 529] and the transformed curve is

$$r^2 + 2\gamma r \cos \theta + m^2 - n^2 = 0,$$

or, as this may be written,

$$r^2 + 2\gamma r \cos \theta + \gamma^2 - f^2 = 0,$$

where $\gamma^2 - f^2 = m^2 - n^2$, that is, $f^2 + m^2 = \gamma^2 + n^2$; viz. this is a concentric circle radius f .

The theorem may be presented as follows. Consider two concentric circles, centre O and radii h, f respectively; take an arbitrary point B , distance $OB = \gamma$; and taking m arbitrary, determine l, n by the equations

$$l^2 = m^2 + h^2 - \gamma^2, \quad n^2 = m^2 + f^2 - \gamma^2;$$

then drawing through B an arbitrary line to meet the circles in A, C respectively; also describing a circle, centre B and radius $= m$; and through A drawing a line ABC perpendicular to BA to meet the last-mentioned circle in two points P, Q : for these points, the distances from the points A, B, C are $= l, m, n$ respectively.

To verify this, take O as the origin, OB for the axis of x , θ the inclination of ABC to this axis, $BA = r'$, $BC = r$; the coordinates of C, B, A are

$$\gamma + r \cos \theta, \quad r \sin \theta,$$

$$\gamma, \quad 0,$$

$$\gamma + r' \cos \theta, \quad -r' \sin \theta,$$

whence, taking (x, y) for the coordinates of P (or Q), the equations to be verified are

$$(x - \gamma - r \cos \theta)^2 + (y - r \sin \theta)^2 = n^2,$$

$$(x - \gamma)^2 + y^2 = m^2,$$

$$(x - \gamma + r' \cos \theta)^2 + (y + r' \sin \theta)^2 = l^2.$$

By means of the second equation, the other two become

$$-2(x - \gamma)r \cos \theta - 2yr \sin \theta + r^2 = n^2 - m^2,$$

$$2(x - \gamma)r' \cos \theta + 2yr' \sin \theta + r'^2 = l^2 - m^2;$$

or, substituting for $n^2 - m^2, l^2 - m^2$ the values $f^2 - \gamma^2$ and $h^2 - \gamma^2$, the equations are

$$-2xr \cos \theta - 2yr \sin \theta + 2\gamma r \cos \theta + r^2 - f^2 + \gamma^2 = 0,$$

$$2xr' \cos \theta + 2yr' \sin \theta - 2\gamma r' \cos \theta + r'^2 - h^2 + \gamma^2 = 0,$$

viz. in virtue of the equations of the two circles, these reduce themselves each of them to

$$x \cos \theta + y \sin \theta = 0,$$

which equation, together with the second equation

$$(x - \gamma)^2 + y^2 = m^2,$$

determine (x, y) as above.

[p. 530] Reverting to the case where the locus of A is the circle

$$r'^2 - 2\gamma r' \cos \theta + \gamma^2 - h^2 = 0,$$

this gives

$$r' = \gamma \cos \theta + \sqrt{(h^2 - \gamma^2 \sin^2 \theta)},$$

$$\frac{1}{r'} = \frac{\gamma \cos \theta - \sqrt{(h^2 - \gamma^2 \sin^2 \theta)}}{\gamma^2 - h^2},$$

so that for the transformed curve we have

$$r^2 + \gamma \cos \theta \left\{ (1 - \lambda)r + (1 + \lambda) \frac{\gamma^2 - h^2}{r} \right\} + (1 - \lambda) \sqrt{(h^2 - \gamma^2 \sin^2 \theta)} + m^2 - n^2 = 0.$$

Putting for shortness $\frac{l^2 - m^2}{\gamma^2 - h^2} = \lambda$, and for $r, r \cos \theta, r \sin \theta$, writing $\sqrt{(x^2 + y^2)}, x, y$ respectively, this is

$$x^2 + y^2 + (1 - \lambda)\gamma x + (1 + \lambda) \sqrt{[h^2(x^2 + y^2) - \gamma^2 y^2]} + m^2 - n^2 = 0,$$

or, what is the same thing,

$$\{x^2 + y^2 + (1 - \lambda)\gamma x + m^2 - n^2\}^2 = (1 + \lambda)^2 \{h^2(x^2 + y^2) - \gamma^2 y^2\},$$

= a bicircular quartic. In the case $\lambda = -1$, it reduces itself to the circle

$$x^2 + y^2 + 2\gamma x + m^2 - n^2 = 0$$

twice, which is the case considered above; and in the case $\lambda = 1$, or $l^2 - m^2 = h^2 - \gamma^2$, the equation is

$$(x^2 + y^2 + m^2 - n^2)^2 = 4\{h^2(x^2 + y^2) - \gamma^2 y^2\},$$

so that the curve is symmetrical in regard to each axis. In the case $\gamma = 0$, the locus is a pair of concentric circles, centre B .

The equation

$$\{x^2 + y^2 + (1 - \lambda)\gamma x + m^2 - n^2\}^2 = (1 + \lambda)^2 \{h^2(x^2 + y^2) - \gamma^2 y^2\},$$

which contains the four constants λ, γ, h and $m^2 - n^2$, may be written in the form

$$(x^2 + y^2 + Ax + B)^2 = ax^2 + ey^2,$$

(where the constants A, B, a, e are also arbitrary). This is, in fact, the equation of the general symmetrical bicircular quartic, referred to a properly-selected point on the axis as origin, viz. the origin is the centre of any one of the three involutions formed by the vertices (or points on the axis); say it is any one of the three involution-centres of the curve.

To show this, assume

$$(x - \alpha)(x - \beta)(x - \gamma)(x - \delta) = x^4 - px^3 + qx^2 - rx + s;$$

[p. 531] then, taking B arbitrary, the equation of the symmetrical bicircular quartic having for vertices the points $x = \alpha, x = \beta, x = \gamma, x = \delta$, is

$$(x^2 + y^2 - \frac{1}{2}px + B)^2 = (2B + \frac{1}{4}p^2 - q)x^2 + (b + 2\theta A)x + (-s + B^2),$$

in fact, this is the form of the general equation, and writing therein $y = 0$, it becomes $x^4 - px^3 + qx^2 - rx + s = 0$, that is, $(x - \alpha)(x - \beta)(x - \gamma)(x - \delta) = 0$. Hence, writing for convenience

$$A = -\frac{1}{2}p,$$

$$a = 2B + \frac{1}{4}p^2 - q,$$

$$b = r - pB,$$

$$c = -s + B^2,$$

the equation is

$$(x^2 + y^2 + Ax + B)^2 = ax^2 + bx + c.$$

This may be written

$$(x^2 + y^2 + Ax + B + \theta)^2 = (a + 2\theta)x^2 + 2\theta y^2 + (b + 2\theta A)x + c + 2B\theta + \theta^2,$$

viz. assuming $\theta = -\frac{b}{2A}$ in order on the right-hand side to destroy the term in x , the equation is

$$\left(x^2 + y^2 + Ax + B - \frac{b}{2A}\right)^2 = \left(a - \frac{b}{A}\right)x^2 + y^2 \cdot \frac{-b}{A} + \frac{1}{4A^2}(b^2 - 4ABb + 4A^2c),$$

which is of the form

$$(x^2 + y^2 + Ax + B)^2 = ax^2 + ey^2 + f;$$

and if $f = 0$, that is, if $b^2 - 4ABb + 4A^2c = 0$, then it is of the required form

$$(x^2 + y^2 + Ax + B)^2 = ax^2 + ey^2.$$

We have

$$\begin{aligned} b^2 - 4ABb + 4A^2c &= (r - pB)^2 + 2pB(r - pB) + p^2(-s + B^2) \\ &= r^2 - p^2s, \end{aligned}$$

or the required condition is $r^2 - p^2s = 0$. But we have

$$r^2 - p^2s = (\alpha\delta - \beta\gamma)(\beta\delta - \gamma\alpha)(\gamma\delta - \alpha\beta),$$

as is easily verified by writing

$$p = \delta + p_r, \quad q = \delta p_r + q_r, \quad r = \delta q_r + r_r, \quad s = \delta r_r,$$

where p_r, q_r, r_r stand for

$$\alpha + \beta + \gamma, \quad \beta\gamma + \gamma\alpha + \alpha\beta, \quad \alpha\beta\gamma,$$

[p. 532] respectively. The required condition thus is

$$(\alpha\delta - \beta\gamma)(\beta\delta - \gamma\alpha)(\gamma\delta - \alpha\beta) = 0,$$

viz. the origin (that is, the fixed point B of the cell) must be at one of the three involution-centres.

Comparing the equation

$$\{x^2 + y^2 + (1 - \lambda)\gamma x + m^2 - n^2\}^2 = (1 + \lambda)^2\{h^2(x^2 + y^2) - \gamma^2 y^2\}$$

with the equation

$$(x^2 + y^2 + Ax + B)^2 = ax^2 + ey^2,$$

we have

$$\begin{aligned} A &= (1 - \lambda)\gamma, \\ B &= m^2 - n^2, \\ a &= (1 + \lambda)^2 h^2, \\ e &= (1 + \lambda)^2 (h^2 - \gamma^2), \end{aligned}$$

and thence $a - e = (1 + \lambda)^2 \gamma^2$. Consequently $\frac{a - e}{A^2} = \left(\frac{1 + \lambda}{1 - \lambda}\right)^2$, which gives λ ; and then

$$h^2 = \frac{a}{(1 + \lambda)^2}, \quad \gamma^2 = \frac{a - e}{(1 + \lambda)^2}, \quad m^2 - n^2 = B;$$

viz. we thus have the values of λ , h , γ and $m^2 - n^2$ for the description of a given curve $(x^2 + y^2 + Ax + B)^2 = ax^2 + ey^2$. In order that the description may be possible, a and $a - e$ must be each of them positive.

For the Cartesian a is $= e$, whence $1 + \lambda = 0$, and the equation becomes

$$(x^2 + y^2 + 2\gamma x + m^2 - n^2)^2 = 0,$$

which is a twice repeated circle; hence the Cartesian cannot be constructed by means of a cell as above.

To obtain a construction of the Cartesian, it may be remarked that, if a symmetrical bicircular quartic be inverted in regard to an axial focus, viz. if the focus be taken as the centre of inversion, we obtain a Cartesian. The axial foci of the curve

$$(x^2 + y^2 + Ax + B)^2 = ax^2 + ey^2$$

are points on the axis, the abscissa $x = \theta$ being determined by the equation

$$e(\theta^2 + A\theta + B)^2 - a(\theta^2 - B)^2 - ae\theta^2 = 0.$$

The equation referred to a focus as origin is therefore

$$\{x^2 + y^2 + (A + 2\theta)x + \theta^2 + A\theta + B\}^2 = ax^2 + ey^2 + 2a\theta x + a\theta^2;$$

then inverting, viz. for x, y writing $\frac{k^2x}{r^2}, \frac{k^2y}{r^2}$ (k arbitrary), we have, as may be verified the equation of a Cartesian.

[p. 533] The inversion can be performed mechanically by an ordinary Peaucellier-cell; the complete apparatus for the construction of a Cartesian is therefore as in fig. 2, viz. we have a cell BAC as before, B a fixed point, locus of A a circle (for convenience of drawing, the arrangement has been made BAC instead of ABC), and we connect with C a Peaucellier-cell $CA'B'$, arms n, n', m' , the fixed point B' being on the axis, which is the line joining B with the centre of the circle described by A . This being so, then A describing a circle, C will describe a symmetrical bicircular quartic, and A' will describe the inverse of this, being in general a like curve; but if the position of B' be properly determined, viz. if B' be at a focus of the first-mentioned quartic.

Fig. 2.

then A' will describe a Cartesian. A further investigation would be necessary in order to determine how to adapt the apparatus to the description of a given Cartesian.

A more convenient mechanical description of a Cartesian is, however, that given in the paper which follows the present one [618].

The equation

$$\{x^2 + y^2 + (1 - \lambda)\gamma x + m^2 - n^2\}^2 = (1 + \lambda)^2\{h^2(x^2 + y^2) - \gamma^2 y^2\}$$

may also be written

$$\begin{aligned} & \{x^2 + y^2 + (1 - \lambda)\gamma x - \frac{1}{2}(1 + \lambda)^2(h^2 - \gamma^2) + m^2 - n^2\}^2 \\ &= (1 + \lambda)^2\{\gamma^2 x^2 - (1 - \lambda)(h^2 - \gamma^2)\gamma x - \frac{1}{4}(1 + \lambda)^2(h^2 - \gamma^2)^2 - (m^2 - n^2)(h^2 - \gamma^2)\}, \end{aligned}$$

viz. the equation is now brought into the form

$$(x^2 + y^2 + Ax + B)^2 = ax^2 + bx + c.$$

Expressing the coefficients A, B, a, b, c in terms of $\lambda, \gamma, h, m^2 - n^2$, it appears by what precedes, that we should have identically $b^2 - 4ABb + 4A^2c = 0$, viz. this is the equation which expresses that the origin is an involution-centre.

If, instead of the original cell, we consider a new cell obtained by substituting for the arms PB, BQ , the arms pb, bq , jointed on to the points p, q on the arms CP, CQ respectively, and instead of B , making

b the fixed point; then writing $Cp = kn$, $Pb = km$, so that the parameters of the cell are l , m , n , k , and taking $Ch = s$, $hA = s'$,

[p. 534] we have $s = kr$, $s + s' = r + r'$, that is, $r = \frac{s}{k}$, $r' = \frac{k-1}{k}s + s'$. Substituting in the equation between r , r' , written for greater convenience in the form

$$(r + r')(rr' + m^2 - l^2) + (l^2 - n^2)r' = 0,$$

the relation between s , s' is found to be

$$(s + s') \left(\frac{k-1}{k^2}s^2 + \frac{ss'}{k} + m^2 - l^2 \right) + (l^2 - n^2) \left(\frac{k-1}{k}s + s' \right) = 0.$$

On account of the term in s^3 , this equation in its general form does not, it would appear, give rise to transformations of much elegance. If, however, $l = n$, then the relation becomes

$$(k-1)s^2 + kss' = k^2(m^2 - l^2);$$

and in particular, if $k = 2$, then

$$s^2 + 2ss' = 4(l^2 - m^2), \quad \text{or say} \quad (s + s')^2 - s'^2 = 4(l^2 - m^2),$$

viz. taking A instead of b as the fixed point, the relation between the radii AC , Ah is $\rho^2 - \rho'^2 = 4(l^2 - m^2)$; the cell is in this case Sylvester's "quadratic-binomial extractor."

618. On the Mechanical Description of a Cartesian

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. *xiii.* (1875), pp. 328–330.]

[p. 535] Suppose that in two different curves the radius vectors r , r' , which belong to the same angle θ , are connected by the equation

$$r^2 + \left(Mr' + N + \frac{P}{r'} \right) r + B = 0;$$

then, taking one of the curves to be the circle

$$Mr' + \frac{P}{r'} = A \cos \theta,$$

the other curve is

$$r^2 + (A \cos \theta + N)r + B = 0,$$

viz. this is a Cartesian. It perhaps would not be difficult to contrive a mechanical arrangement to connect the radius vectors in accordance with the foregoing equation; but the required result may be obtained equally well by means of a particular case of the relation in question; viz. taking this to be

$$r^2 + (-r' + N)r + B = 0,$$

then, taking the one curve to be the circle $r' = A \cos \theta$, the other curve is the Cartesian,

$$r^2 + (-A \cos \theta + N)r + B = 0, \quad \text{that is,} \quad r^2 + (A \cos \theta + N)r + B = 0.$$

The relation between the radius vectors may in this case be written

$$r' = N + r + \frac{B}{r},$$

which can be constructed mechanically by a simple addition to the Peaucellier-cell, viz. if we joint on to C (fig. 1, p. 536) a rod CD having a slot, working on a pin at A , so that the rod is thereby kept always in the line BAC , then, making B the

[p. 536] fixed point and taking $BA = r$, we have $AC = \frac{B}{r}$, whence $BC = r + \frac{B}{r}$, or D being a point at the distance $CD = a$, from the point C , and denoting BD by r' , we have

$$r' = r + \frac{B}{r} + a,$$

which is an equation of the required form; whence, if the point A describe a circle passing through B , then the point D will describe a Cartesian.

Fig. 1.

The equation of the Cartesian is $r^2 + (A \cos \theta + N)r + B = 0$, viz. this is

$$x^2 + y^2 + Ax + B = -N\sqrt{(x^2 + y^2)},$$

or writing $N^2 = a$, it is

$$(x^2 + y^2 + Ax + B)^2 = ax^2 + ay^2,$$

which is the form considered in the preceding paper. It may be further observed in regard to it that, starting from the focal equation $r = ls + m$, where r, s are the distances of a point (x, y) of the Cartesian from any two of its three foci, this equation gives $r^2 - l^2 s^2 = m^2 + 2mr$, or writing $r^2 = x^2 + y^2$, $s^2 = (x - a)^2 + y^2$, the function on the left-hand is of the form $(1 + l^2)(x^2 + y^2 + Ax + B)$, whence, assuming $\sqrt{(1 + l^2)} = \sqrt{(a)}$, the equation becomes as above

$$(x^2 + y^2 + Ax + B)^2 = a(x^2 + y^2).$$

Taking the distance $r = \sqrt{(x^2 + y^2)}$ to be measured from a given focus, it is easy to see that, no matter which of the other two foci we associate with it, we obtain the same equation $(x^2 + y^2 + Ax + B)^2 = a(x^2 + y^2)$; viz. starting with any one focus, we connect with it a determinate circle $x^2 + y^2 + Ax + B = 0$, and a determinate coefficient a , such that taking this focus as the origin, the equation of the curve is

$$(x^2 + y^2 + Ax + B)^2 = a(x^2 + y^2);$$

but there are for the given curve three such forms of equation, according as the origin is taken at one or other of the three foci.

(*Addition, Feb. 1875.*) It is obviously the same thing, but I find that it is mechanically more convenient to derive the Cartesian from the Limaçon $r' = N - A \cos \theta$, by the transformation $r' = r + \frac{B}{r}$: I have on this principle constructed an apparatus whereby the Cartesian is described on a rotating board by a pencil moving in a fixed line.

619. On an Algebraical Operation

[*From the Quarterly Journal of Pure and Applied Mathematics, vol. xiii. (1875), pp. 369–375.*]

[p. 537] I consider

$$\Omega F(a, x),$$

an operation Ω performed upon $F(a, x)$ a rational function of (a, x) ; viz. F being first expanded or regarded as expanded in ascending powers of a , the coefficients of the several powers are then to be expanded or regarded as expanded in ascending powers of x , and the operation consists in the rejection of all negative powers of x .

In the cases intended to be considered, F contains only positive powers of a : but this restriction is not necessary to the theory.

The investigation has reference to the functions $A(x)$ of my “Ninth Memoir on Quantics,” *Phil. Trans.*, t. CLXI. (1871), pp. 17–50, [462]; for instance, we there have as regards the covariants of a quadric

$$A(x) = \frac{1 - x^{-4}}{1 - ax^2 \cdot 1 - a \cdot 1 - ax^{-2}},$$

and consequently, in the present notation,

$$A(x) = \Omega \frac{1 - x^{-4}}{1 - ax^2 \cdot 1 - a \cdot 1 - ax^{-2}};$$

by a process of development and summation, the value of this expression was found to be

$$\frac{1}{1 - ax^2 \cdot 1 - a^2},$$

[p. 538] and in the other more complicated cases the value of $A(x)$ was found only by trial and verification. What I purpose now to show is that the operation Ω can be performed without any development in an infinite series; or say that it depends on finite algebraical operations only.

It is clear that if $F(a, x)$, considered to be developed as above contains only positive powers of x , then

$$\Omega F(a, x) = F(a, x);$$

And if it contains only negative powers of x , then $\Omega F(a, x) = 0$.

Consider now $\Omega \frac{\phi(x)}{x - a}$, where $\phi(x)$ is a function containing only positive powers of x ; we have

$$\frac{\phi(x)}{x - a} = \frac{\phi(x) - \phi(a)}{x - a} + \frac{\phi(a)}{x - a},$$

and thence

$$\Omega \frac{\phi(x)}{x - a} = \Omega \frac{\phi(x) - \phi(a)}{x - a} + \Omega \frac{\phi(a)}{x - a} = \frac{\phi(x) - \phi(a)}{x - a},$$

since $\frac{\phi(x) - \phi(a)}{x - a}$ is a rational and integral function of x , which when developed contains only positive powers of x , and $\frac{\phi(a)}{x - a}$ when developed contains only negative powers of x .

Consider next $\Omega \frac{\phi(x)}{x^2 - a}$, where $\phi(x)$ is a rational and integral function of x ; writing this = $f(x^2) + xg(x^2)$, we have

$$\Omega \frac{\phi(x)}{x^2 - a} = \Omega \frac{f(x^2)}{x^2 - a} + \Omega \frac{xg(x^2)}{x^2 - a} = \frac{f(x^2) - f(a)}{x^2 - a} + \Omega \frac{xg(x^2)}{x^2 - a}.$$

As regards the last term, notice that

$$\frac{xg(x^2)}{x^2 - a} = \frac{x[g(x^2) - g(a)]}{x^2 - a} + \frac{xg(a)}{x^2 - a},$$

in which $\frac{x[g(x^2) - g(a)]}{x^2 - a}$ is a rational and integral function of (a, x) , and therefore when developed contains only positive powers of x , while $\frac{xg(a)}{x^2 - a}$ when developed contains only negative powers of x .

We thus have

$$\Omega \frac{\phi(x)}{x^2 - a} = \frac{f(x^2) + xg(x^2) - f(a) - xg(a)}{x^2 - a} = \frac{\phi(x) - f(a) - xg(a)}{x^2 - a}.$$

[p. 539] Similarly, if $\phi(x) = f(x^3) + xg(x^3) + x^2h(x^3)$, then

$$\Omega \frac{\phi(x)}{x^3 - a} = \frac{\phi(x) - f(a) - xg(a) - x^2h(a)}{x^3 - a},$$

and so on.

Consider now the above-mentioned function

$$A(x) = \Omega \frac{1 - x^{-2}}{1 - ax^2 \cdot 1 - a \cdot 1 - ax^{-2}}.$$

Writing

$$\frac{1 - x^{-2}}{1 - ax^2 \cdot 1 - a \cdot 1 - ax^{-2}} = \frac{P}{1 - ax^2} + \frac{Q}{1 - a} + \frac{R}{1 - ax^{-2}},$$

we have

$$\begin{aligned} P &= \left(\frac{1 - x^{-2}}{1 - a \cdot 1 - ax^{-2}} \right)_{a=x^{-2}} = \frac{1}{1 - x^{-4}} = \frac{-x^4}{1 - x^4}, \\ Q &= \left(\frac{1 - x^{-2}}{1 - ax^2 \cdot 1 - ax^{-2}} \right)_{a=1} = \frac{1}{1 - x^2}, \\ R &= \left(\frac{1 - x^{-2}}{1 - ax^2 \cdot 1 - a} \right)_{a=x^2} = \frac{1 - x^{-2}}{1 - x^4 \cdot 1 - x^2} = \frac{-1}{x^2(1 - x^4)}; \end{aligned}$$

that is,

$$\frac{1 - x^{-2}}{1 - ax^2 \cdot 1 - a \cdot 1 - ax^{-2}} = \frac{-x^4}{1 - x^4} \cdot \frac{1}{1 - ax^2} + \frac{1}{1 - x^2} \cdot \frac{1}{1 - a} - \frac{1}{1 - x^4} \cdot \frac{1}{x^2 - a},$$

and thence

$$\Omega \frac{1 - x^{-2}}{1 - ax^2 \cdot 1 - a \cdot 1 - ax^{-2}} = \frac{-x^4}{1 - x^4} \cdot \frac{1}{1 - ax^2} + \frac{1}{1 - x^2} \cdot \frac{1}{1 - a} + \Omega \frac{-1}{1 - x^4} \cdot \frac{1}{x^2 - a}.$$

Here, as regards the last term,

$$\phi(x^2) = \frac{-1}{1 - x^4} = f(x^2), \quad g(x^2) = 0,$$

$$f(a) = \frac{-1}{1 - a^2},$$

and

$$\frac{\phi(x^2) - f(a)}{x^2 - a} = \frac{1}{x^2 - a} \left(\frac{-1}{1 - x^4} + \frac{1}{1 - a^2} \right) = -\frac{x^2 + a}{1 - x^4 \cdot 1 - a^2},$$

and we have

$$A(x) = \frac{-x^4}{1 - x^4} \cdot \frac{1}{1 - ax^2} + \frac{1}{1 - x^2} \cdot \frac{1}{1 - a} - \frac{x^2 + a}{1 - x^4 \cdot 1 - a^2}.$$

The second term is $= \frac{(1 + x^2)(1 + a)}{(1 - x^4)(1 - a^2)}$; combining this with the third term, the two together are $= \frac{1 + ax^2}{1 - x^4 \cdot 1 - a^2}$.

[p. 540] Hence the value is

$$= \frac{1}{1 - x^4} \left(\frac{-x^4}{1 - ax^2} + \frac{1 + ax^2}{1 - a^2} \right),$$

which is

$$= \frac{1}{1 - ax^2 \cdot 1 - a^2},$$

being, in fact, the expression for this function when decomposed into partial fractions of the denominators $1 - ax^2$ and $1 - a^2$ respectively. Hence finally

$$A(x) = \Omega \frac{1 - x^{-2}}{1 - ax^2 \cdot 1 - a \cdot 1 - ax^{-2}} = \frac{1}{1 - ax^2 \cdot 1 - a^2},$$

as it should be.

For the cubic function, we have

$$A(x) = \Omega \frac{1 - x^{-2}}{1 - ax^3 \cdot 1 - ax \cdot 1 - ax^{-1} \cdot 1 - ax^{-3}};$$

the function operated upon, when decomposed into partial fractions, is

$$= \frac{x^{10}}{1 - x^4 \cdot 1 - x^6} \cdot \frac{1}{1 - ax^3} - \frac{x^4}{1 - x^2 \cdot 1 - x^4} \cdot \frac{1}{1 - ax}$$

$$+\frac{x}{1-x^2 \cdot 1-x^4} \cdot \frac{1}{x-a} + \frac{-x}{1-x^4 \cdot 1-x^6} \cdot \frac{1}{x^3-a}.$$

Hence we require

$$\Omega \frac{x}{1-x^2 \cdot 1-x^4} \cdot \frac{1}{x-a} + \Omega \frac{-x}{1-x^4 \cdot 1-x^6} \cdot \frac{1}{x^3-a}.$$

The first of these is

$$= \frac{1}{x-a} \left(\frac{x}{1-x^2 \cdot 1-x^4} - \frac{a}{1-a^2 \cdot 1-a^4} \right),$$

which is

$$\frac{1}{1-x^2 \cdot 1-x^4 \cdot 1-a^2 \cdot 1-a^4} \left\{ \begin{array}{l} 1 \\ +x(a+a^3-a^5) \\ +x^2(a^2-a^4) \\ +x^3(a-a^3) \\ -x^4a^2 \\ -x^5a \end{array} \right\}.$$

As regards the second, the function operated on may be expressed in the form

$$\frac{-x^9-x-x^5}{1-x^4 \cdot 1-x^6} \cdot \frac{1}{x^3-a},$$